

Group02 - PA2 - Use Case Document

Project Information

- **Course:** Human - Computer Interaction
- **Topic:** Running Music Coach - Adaptive Music and Eyes-Free Interaction for Runners
- **Team:** Group02
- **Lecturer:** Le Khanh Duy
- **Date:** July 2026

Members

Student ID	Full name
23127006	Trần Nguyễn Khải Luân
23127179	Nguyễn Bảo Duy
23127271	Võ Ngọc Bích Trâm
23127372	Mai Xuân Hưng

1. System Overview

System name: Running Music Coach

Purpose: Support runners with adaptive music, low-attention pace feedback, and reliable in-run controls.

2. Actors and Artifacts

Primary Actor

- Runner

Secondary Actors

- Music Service
- Running Tracker
- Notification Service
- Smartwatch / Wearable Device
- Earbuds

Main Artifacts

- phone app interface

- watch interface
- music playback channel
- haptic feedback channel
- notification filter / Focus Run Mode
- running session data and summary

3. Preconditions

- User has installed and opened the system.
- Basic permissions for activity tracking, audio, and notifications are granted.
- A playlist or audio source is available.
- If used, smartwatch or earbuds are connected.

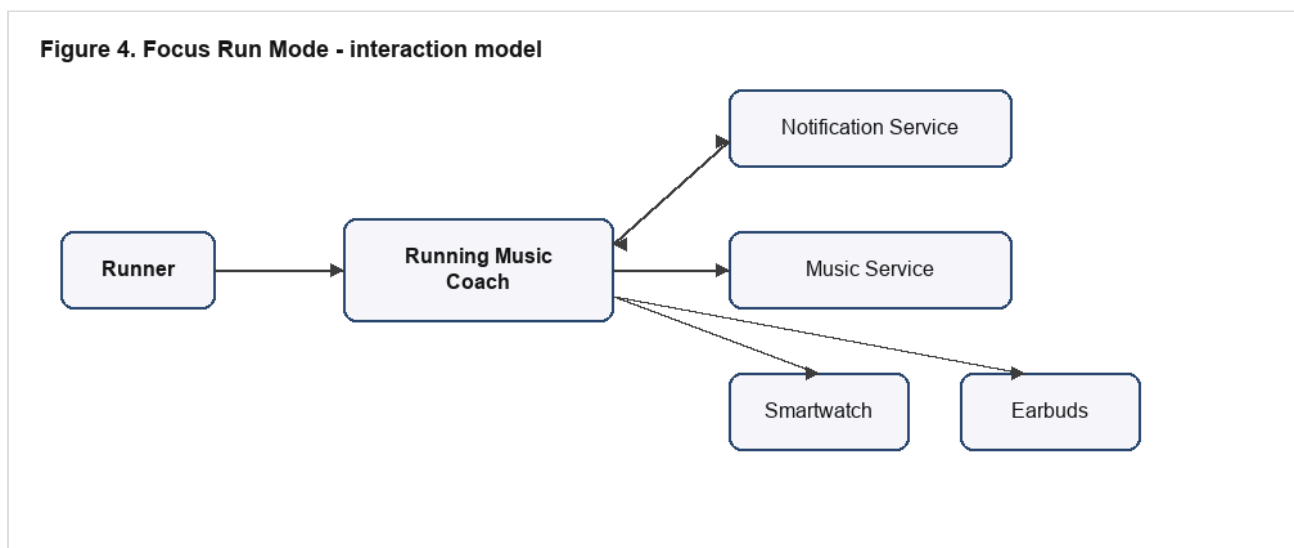
4. Conceptual Solutions and Their Interaction Models

The proposal document selected several conceptual solutions. This use case document provides a flow model and use case specifications for those solutions.

4.1 Conceptual Solution A - Focus Run Mode

Purpose: Protect music flow by handling interruptions differently during an active run.

Figure 4. Focus Run Mode interaction model

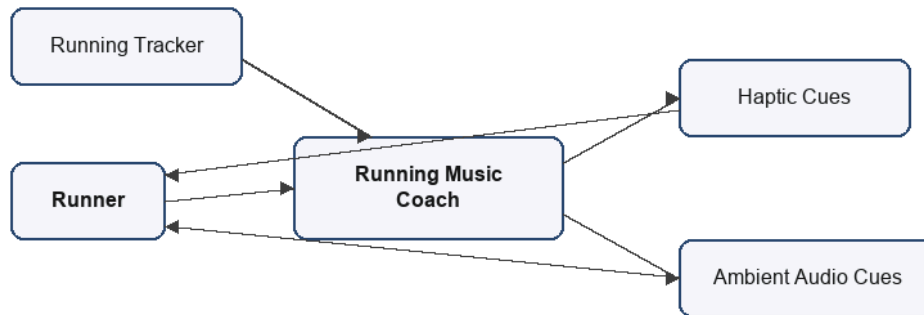


4.2 Conceptual Solution B - Eyes-Free Pace Feedback

Purpose: Help runners understand pace/effort state without repeated screen checks.

Figure 5. Eyes-Free Pace Feedback interaction model

Figure 5. Eyes-Free Pace Feedback - interaction model

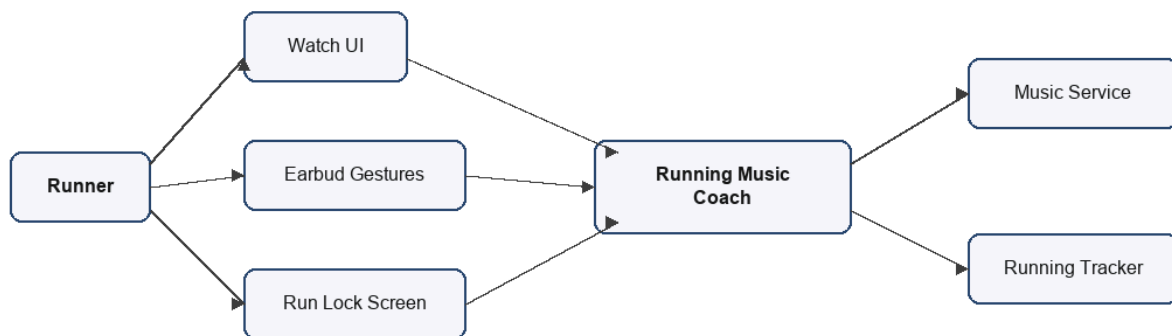


4.3 Conceptual Solution C - Reliable Quick Controls

Purpose: Reduce pause/resume/control friction during movement.

Figure 6. Reliable Quick Controls interaction model

Figure 6. Reliable Quick Controls - interaction model



5. Use Case List

ID	Use Case	Related conceptual solution
UC-01	Start Focused Running Session	A
UC-02	Filter Non-Critical Interruptions	A
UC-03	Maintain Pace via Eyes-Free Feedback	B
UC-04	Recover from Overexertion Signal	B
UC-05	Perform Quick In-Run Controls	C
UC-06	End Session and Review Summary	A, B, C

6. Detailed Use Case Specifications

UC-01 - Start Focused Running Session

- **Primary actor:** Runner
- **Goal:** Begin a run with tracking, music, and interruption rules enabled
- **Trigger:** Runner chooses to start a run
- **Preconditions:** App is open; playlist and permissions are available

Main flow

1. Runner opens Running Music Coach.
2. Runner chooses route, goal, or run mode.
3. Runner selects or confirms music source.
4. System enables Focus Run Mode according to saved preferences.
5. System starts activity tracking.
6. System begins music playback.
7. System enters active monitoring state.

Alternative flows

- A1. If no playlist is selected, the system suggests recent or recommended playlists.
- A2. If a required permission is missing, the system asks for permission before continuing.

Postconditions

- Run session is active.
- Music and tracking are running together.

UC-02 - Filter Non-Critical Interruptions

- **Primary actor:** Runner
- **Secondary actor:** Notification Service
- **Goal:** Prevent non-critical interruptions from breaking running flow
- **Trigger:** A call, message, or notification arrives during a run
- **Preconditions:** Focus Run Mode is active

Main flow

1. Notification Service sends an interruption event.
2. System checks the interruption against the user policy.
3. If the interruption is non-critical, the system suppresses or defers it.
4. Music continues without spoken cut-in.
5. The system stores the deferred event for later review.

Alternative flows

- A1. If the event is marked critical, the system allows it through.
- A2. If the user has whitelisted the sender, the system allows the interruption.

Postconditions

- Music continuity is preserved for non-critical events.

UC-03 - Maintain Pace via Eyes-Free Feedback

- **Primary actor:** Runner
- **Secondary actor:** Running Tracker
- **Goal:** Help the runner know pace or effort state without repeatedly looking at a screen
- **Trigger:** Sustained deviation from target pace, cadence, or effort
- **Preconditions:** Run session is active and pace monitoring is enabled

Main flow

1. Running Tracker sends pace, cadence, and optional heart-rate data.
2. System compares current state with the target zone.
3. System waits for a short stability threshold to avoid noisy alerts.
4. If deviation persists, the system emits a subtle haptic or ambient cue.
5. Runner adjusts speed or stride.
6. System returns to monitoring.

Alternative flows

- A1. If no smartwatch is available, the system uses ambient audio only.
- A2. If the user disables sound cues, the system uses haptic cues only.

Postconditions

- Runner receives pace feedback with low attention cost.

UC-04 - Recover from Overexertion Signal

- **Primary actor:** Runner
- **Goal:** Help the runner stabilize after signs of sustained overexertion
- **Trigger:** System detects pace/effort pattern above the desired threshold
- **Preconditions:** Session is active and overexertion logic is enabled

Main flow

1. System detects a sustained overexertion pattern.
2. System issues a gentle slow-down cue.
3. System may recommend or shift to a less intense track profile.
4. Runner reduces pace and stabilizes breathing.
5. System confirms return to an acceptable zone.

Postconditions

- Runner can recover without an abrupt breakdown in session flow.

UC-05 - Perform Quick In-Run Controls

- **Primary actor:** Runner
- **Goal:** Execute core actions quickly and reliably while moving
- **Trigger:** Runner uses watch, earbud gesture, or Run Lock screen
- **Preconditions:** Session is active

Main flow

1. Runner performs a quick action such as pause, resume, skip track, or check status.
2. The input is received from watch, earbuds, or Run Lock screen.
3. System confirms and executes the action.
4. System returns to active monitoring state.

Alternative flows

- A1. If the input is ambiguous, the system requests a simple confirmation.
- A2. If earbud gestures are unavailable, the system falls back to watch or large-button phone UI.

Postconditions

- Core action is completed with minimal interaction overhead.

UC-06 - End Session and Review Summary

- **Primary actor:** Runner
- **Goal:** End the run and review key session outcomes
- **Trigger:** Runner selects End Run
- **Preconditions:** Session is active

Main flow

1. Runner ends the session.
2. System stops tracking and finalizes activity data.
3. System shows a concise summary including time, distance, pace trend, and notable cue events.
4. System syncs data to connected services.

Alternative flows

- A1. If sync fails, the system stores the result locally and retries later.

Postconditions

- Session result is saved and available for review.

7. Non-Functional Requirements

- **Responsiveness:** In-run quick controls should respond immediately.
- **Reliability:** Core actions should remain usable under sweaty and high-motion conditions.
- **Low attention cost:** Feedback should avoid demanding continuous visual focus.
- **Safety-aware behavior:** Critical alerts must still be possible when required.

- **Data trust:** Sync mismatch should be surfaced clearly instead of silently ignored.

8. Traceability to Research Problems

Research problem	Supporting use cases
Music interruptions reduce motivation	UC-01, UC-02
Visual pace checks disrupt flow	UC-03, UC-04
Touch interaction is error-prone mid-run	UC-05

9. Conclusion

These use cases translate the selected conceptual solutions into concrete interaction specifications. Together, they show the actors, artifacts, and interactions required for a running experience that protects music flow, reduces visual dependence, and lowers in-run interaction friction.